

NEXUS: Advanced Static Analysis & Threat Intelligence



Technical Memo

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Overview

NEXUS is a multi-vector malware detection platform developed by Team Opulence at Prairie View A&M University in partnership with the National Security Agency. The platform addresses critical gaps in existing threat detection tools by combining speed, transparency, and privacy-first architecture.

Why Threat Analysis Matters – AT&T Data Breach

In March of 2024, AT&T experienced a data breach leading to 73 million customer records being exposed. This breach not only cost \$13 million in direct losses, but immeasurable damage to their reputation. Figure 1 details the timeline of this event.

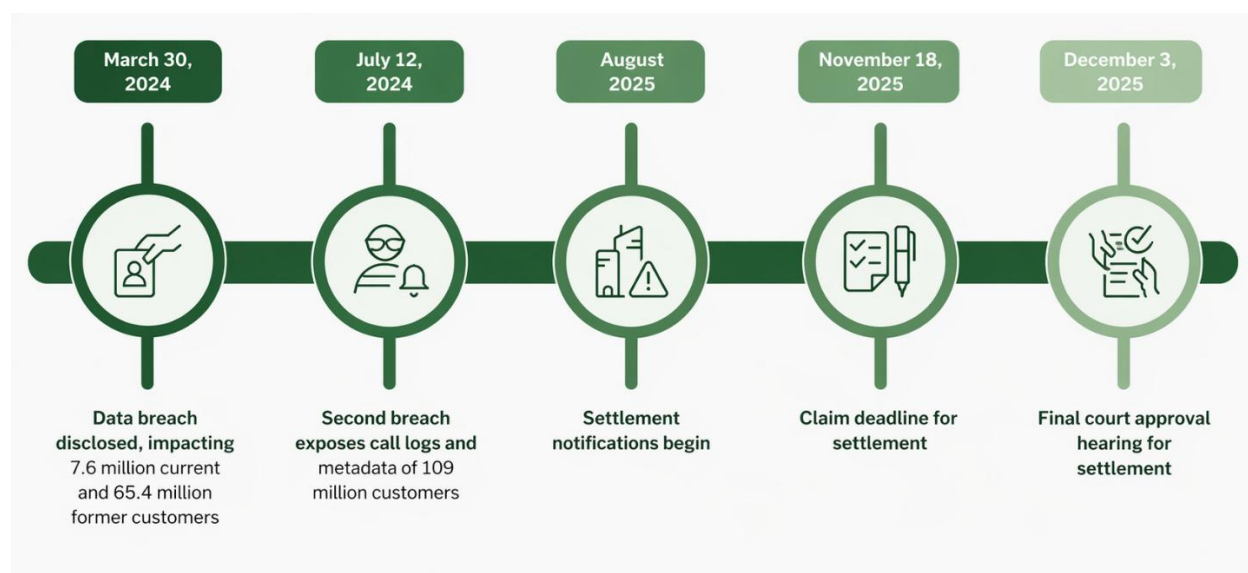


Figure 1. AT&T Data Breach Timeline

The data breach timeline begins in March 2024, with the initial disclosure and concludes in December 2025. Following the first breach, a second breach exposing 109 million customer records occurs in July of 2024. Once the breaches concluded, settlement notifications began August 2025, and final court approval was granted in December 2025. This timeline illustrates the time that it takes a large company to fully resolve, and issue like a data breach, over a year and a half. The initial cause of this breach, a suspicious file or deceptive link that bypassed initial defenses.

Industry statistics quantify the cybersecurity gap with these four core challenges: fast automated tools may miss sophisticated threats, thorough analysis takes too long, existing solutions lack transparency, and privacy concerns persist with third-party uploads.

Problem Statement

Organizations face an escalating cybersecurity crisis. The average data breach costs an organization on average \$4.45M (IBM, 2023). Manual analysis of a single suspicious file can take upwards of 2 to 4 hours. Additionally, 92% of organizations that on signature-based detection and only. 45% of malware is delivered through files, links, and phishing emails, which are all common in an office setting.

Current tools like AVG and VirusTotal each address only a portion of the problem. Table 1 provides information regarding two popular softwares.

Table 1. Comparison of Current Tools

Software	Details
AVG	✓ Fast signature-based scanning ✗ Limited explainability ✗ May miss new or obfuscated threats ✗ Limited metadata/context analysis
VirusTotal	✓ Useful quick triage ✗ Limited control / privacy ✗ third-party file submission concerns ✗ external dependency

Many of these existing tools require tradeoffs between speed, transparency, privacy, or deployment control.

Our Solution: NEXUS

Our solution is NEXUS, software which addresses every limitation by delivering transparent, explainable results with local processing, fast and accurate analysis (<8 sec, 90% accuracy), and three distinct analysis methods. Table 2 provides a Side-by-side comparison of NEXUS versus other tools across four dimensions.

Table 2. Side-By-Side Comparison to NEXUS

Software	Results	Privacy	Speed	Methods
Other Tools	Often limited explanation	May require external services	Often tradeoffs in speed or depth	Limited workflow flexibility
NEXUS	Why it's malicious (ips, URLs, metadata)	Local processing - Files never stored	Fast and accurate (<8 sec, 90% accuracy)	File, URL, email (3 analysis methods)

System Architecture & Backend

The NEXUS platform follows a modular architecture connecting the React.js frontend to the Flask/Python backend through a multi-component analysis engine and into Snowflake cloud storage. Figure 2 depicts the end-to-end system architecture frontend through results dashboard.

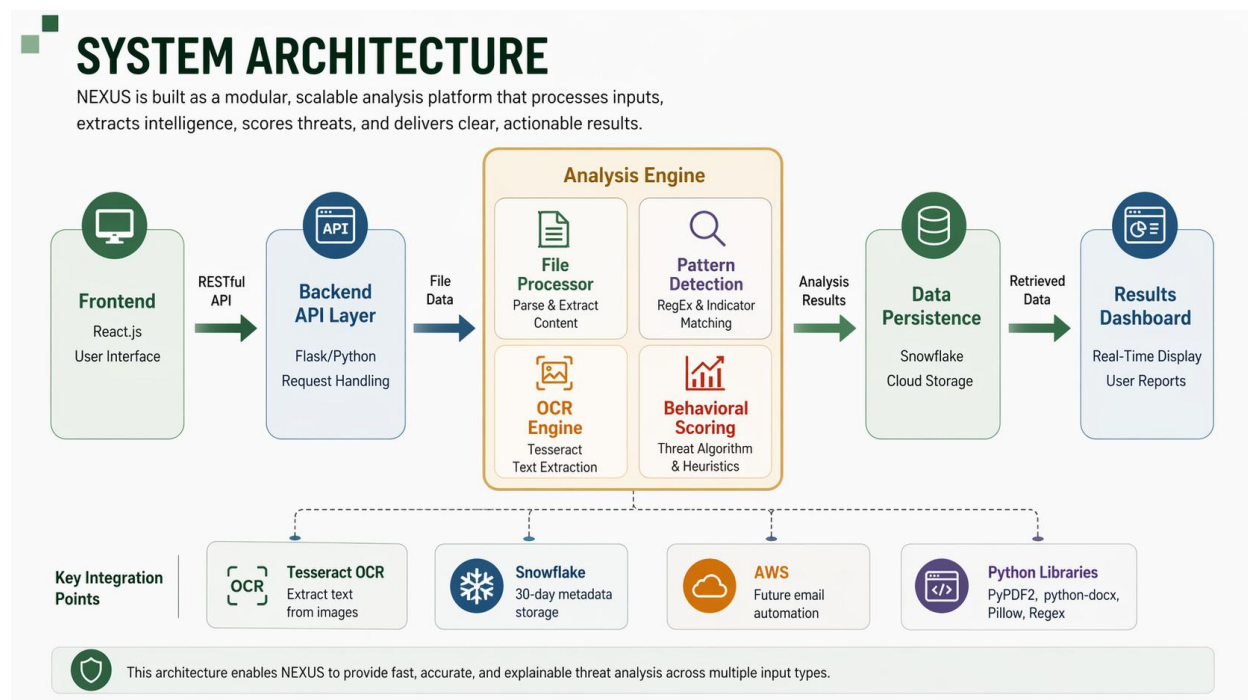


Figure 2. End-to-End System Architecture

Figure 3 details why each indicator matters as, the NEXUS platform follows a four-step process: Detect, Understand, Explain, and Educate & Protect.

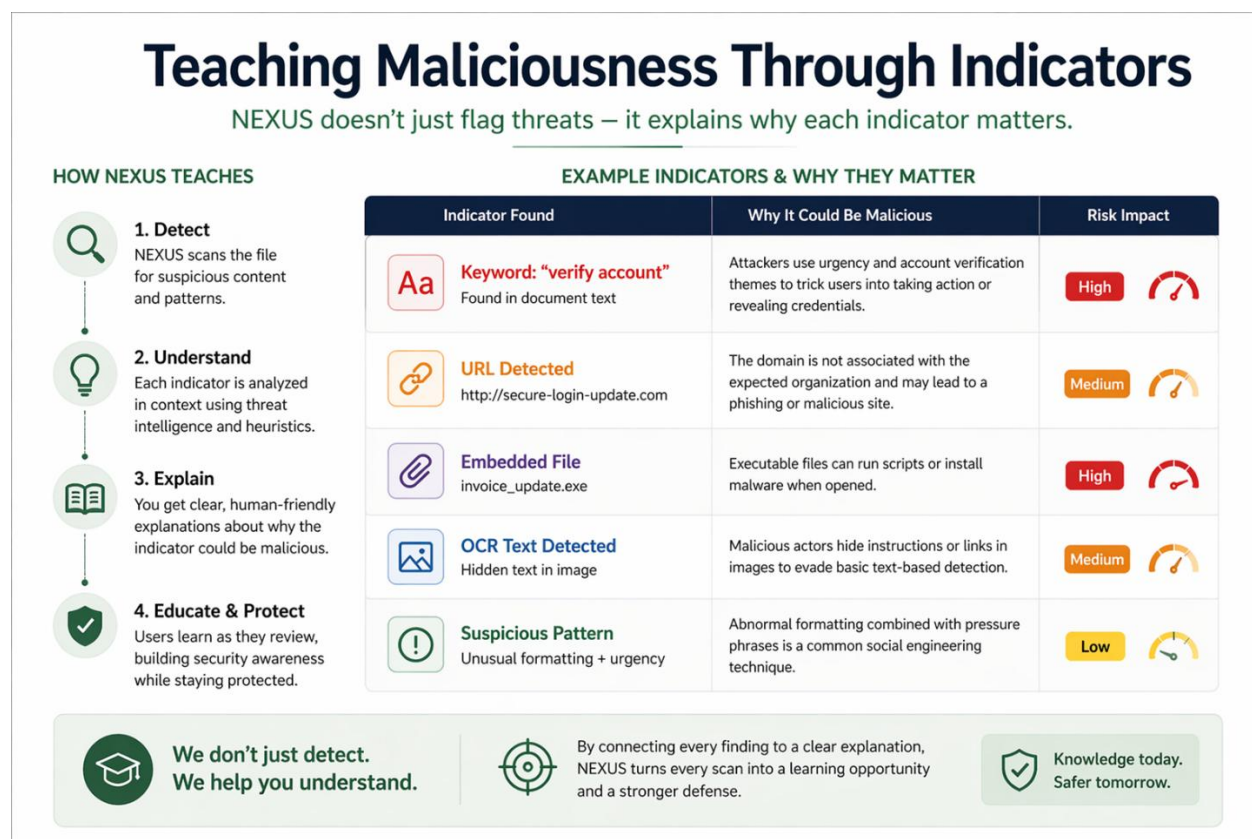


Figure 3. Indicators and Why They Matter

The backend pipeline, depicted in Figure 4, analyzes suspicious inputs through five stages and turns them into fast, explainable security decisions. The 5-step backend pipeline is as follows:

1. Intake – receives uploaded files, URLs, or .eml emails
2. Extraction – pulls text, metadata, hashes, and OCR content
3. Detection – searches for suspicious indicators using patterns and threat intelligence
4. Scoring – assigns risk levels (Clean, Warning, Malicious)
5. Reporting – displays explainable results and stores scan metadata.

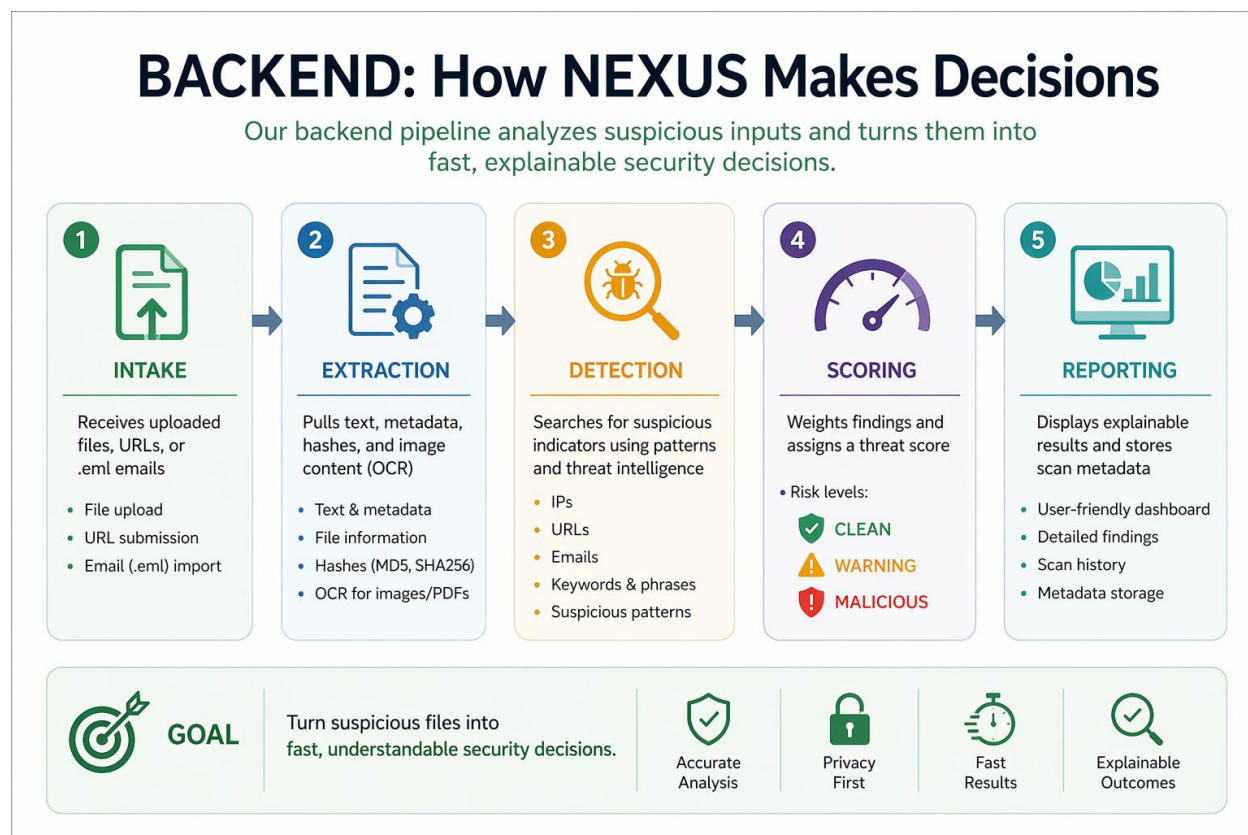


Figure 4. Depiction of Backend Process

Frontend Design & User Experience

The NEXUS frontend was designed around four pillars: Simplicity (3-click maximum), Clarity (color-coded severity), Accessibility (WCAG 2.1 AA), and Speed (real-time progress indicators). Table 3 provides details regarding these pillars.

Table 3. Frontend Pillars of NEXUS

Simplicity	Accessibility	Clarity	Speed
3-click maximum to results - Minimal cognitive load - No cybersecurity expertise required	- WCAG 2.1 AA compliance - Screen reader compatible - Keyboard navigation support - High contrast ratios	- Visual threat scoring - Color-coded severity (Green/Amber/Red) - Plain language explanations	- Real-time progress indicators - Instant feedback - Non-blocking operations

Our Design Approach Is: “Enterprise-grade security should feel consumer-simple.”

Our frontend technical design is organized into three categories: Technology, Features, and Accessibility. Details on each category are provided in Table 4. It is built with React.js, responsive CSS, RESTful APIs, and WebSocket updates for real-time communication. Features include drag-and-drop upload, threat visualization, and dark/light mode.

Table 4. Technical Design of Frontend

Technology	Features	Accessibility
- React.js - Responsive CSS - RESTful API - WebSocket Updates	- Drag & Drop Upload - Real-Time Progress - Threat Visualization - Export Reports	- WCAG 2.1 AA - Screen Readers - Keyboard Nav - Dark/Light Mode

A real screenshot of the NEXUS dashboard is provided in Figure 5. This screenshot depicts a malicious .txt file scan with a threat score of 100, detailed suspicious indicators, and plain-language explanations for each finding.

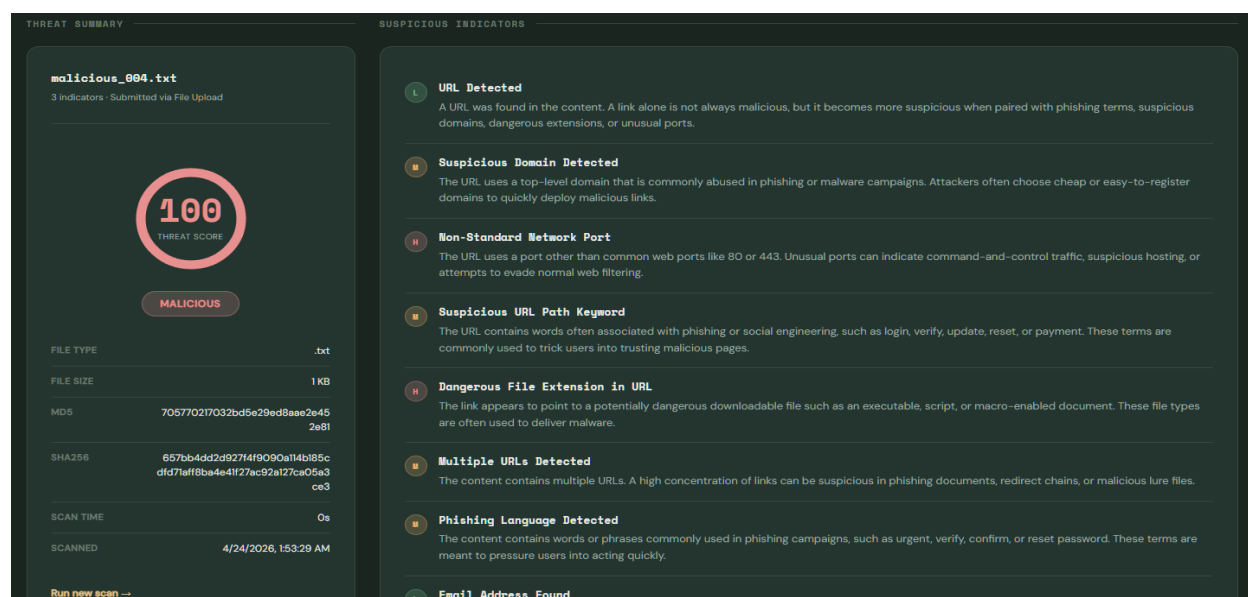


Figure 5. Screenshot Of the NEXUS Dashboard

The left panel displays the threat summary with a score of 100 (MALICIOUS), file metadata, and hash values. The right panel lists each suspicious indicator found—URLs, suspicious domains, non-standard ports, phishing language—with clear explanations of why each one matters.

Testing & Performance Results

NEXUS was evaluated using a balanced dataset of 1,000 files (500 clean, 500 malicious) distributed equally across four file types: .DOCX, .PDF, .TXT, and .PNG (250 each). This process was done to ensure a balanced evaluation, format diversity, robust validation, and reliable results.

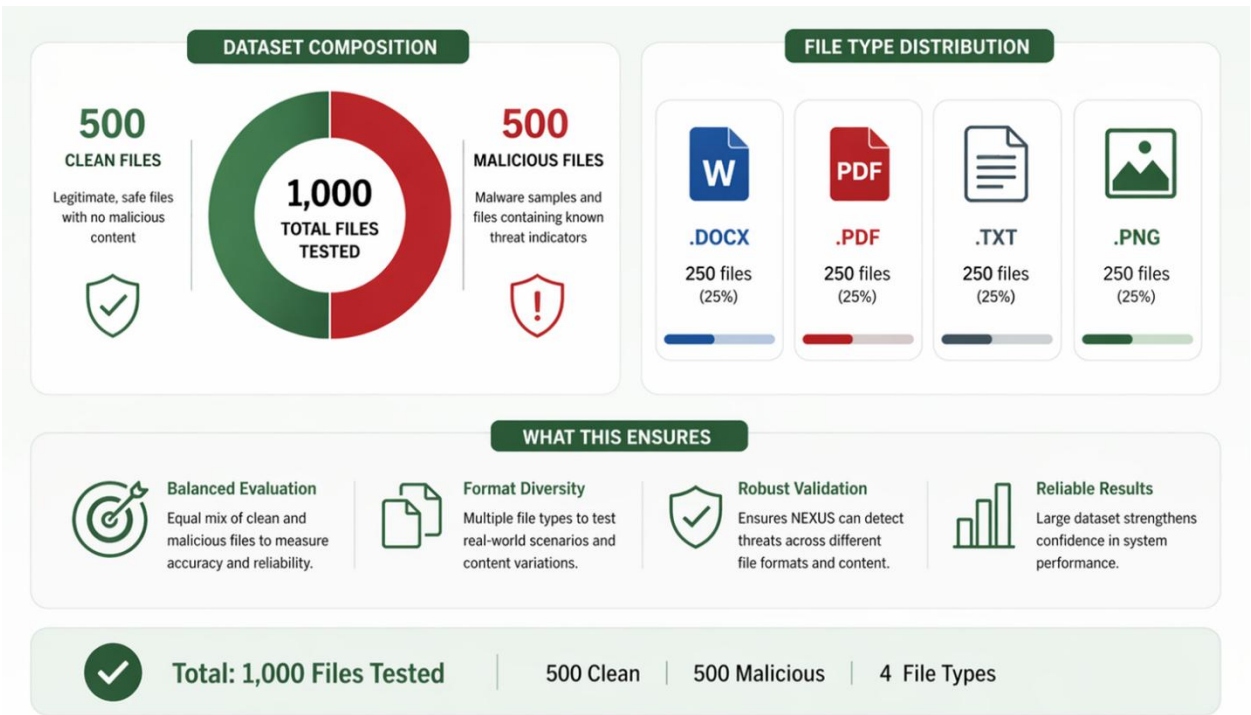
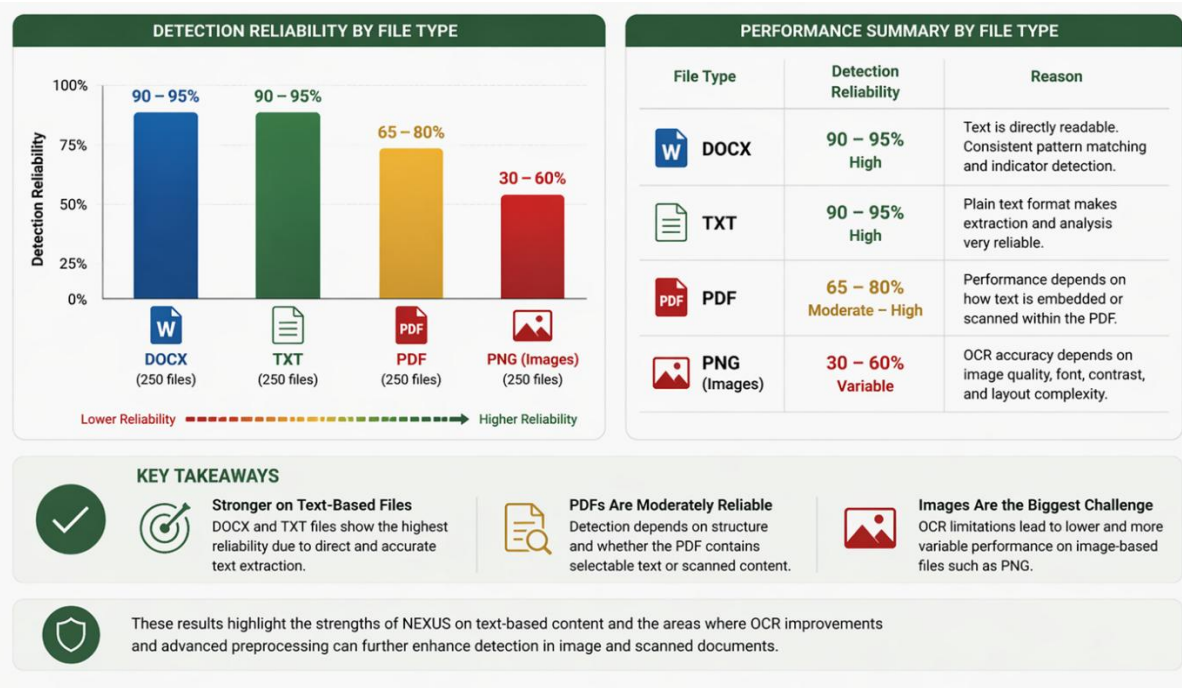


Figure 6. Testing Dataset Overview

Detection reliability varies by file format. Text-based files (DOCX, TXT) achieve 90–95% reliability, PDFs reach 65–80% depending on structure, and PNG images range 30–60% due to OCR variability. Figure 7 provides a detection reliability chart and table by file type.



1. INDIVIDUAL INDICATOR WEIGHTS	
INDICATOR	WEIGHT (POINTS)
 URL Detected	+5
 Multiple URLs (2 or more)	+10
 Email Address Found	+5
 Phishing Language / Keywords (e.g., urgent, verify, account, login)	+15
 Suspicious Domain (low reputation / blacklisted)	+20
 Dangerous File Extension in URL (e.g., .exe, .zip, .scr, .js)	+20
 Non-Standard or Suspicious Port (e.g., :8080, :4444, :8888)	+15

2. COMBINATION BONUSES	
COMBINED INDICATORS	BONUS (POINTS)
 +  URL + Email (appears together)	+15
 +  URL + Phishing Language (appears together)	+20
 +  +  Multiple Indicator Types (3 or more different types)	+25



SCORING NOTES

- Points from all triggered indicators are added together.
- Combination bonuses are applied on top of individual weights.
- Final score is capped at 100 points.

3. FINAL SCORE INTERPRETATION (0–100)		
 0 – 30 CLEAN / LOW RISK No significant threats detected.	 31 – 69 WARNING / MODERATE RISK Suspicious content detected. Exercise caution.	 70 – 100 MALICIOUS / HIGH RISK Highly suspicious content. Immediate attention recommended.



This scoring model is transparent, explainable, and easy to tune as new threat intelligence and patterns are discovered.

Figure 8. Breakdown of Backend Threat Score Logic

Live Demonstration

During the presentation, we demonstrate all three analysis methods and walk through the results interpretation workflow. Table 5 provides the details for each step of the methodology.

Table 5. Live Demo Plan

Analysis Method	What We'll Demonstrate
File Upload (Primary Demo)	- Upload malicious PDF/TXT sample - Upload PNG image Sample - Threat score + Indicator Explanations
URL Analysis (Brief Demo)	- Submit Sample URL - Indicator Review - Risk scoring workflow
Email File (.eml) Analysis (Brief Demo)	- Upload .eml sample - Sender/body extraction - Phishing indicator detection

Results Dashboard

- Threat score explanation
- Show why item was flagged
- User-friendly findings display

Conclusion

NEXUS demonstrates how accessible, explainable security tools can help users identify suspicious content more efficiently. The platform detects suspicious indicators in files, URLs, and email samples, uses OCR for image-based content, provides threat scores with clear explanations, and helps users make faster, smarter security decisions.

Built in partnership with the National Security Agency and validated against 1,000 real-world samples across four file formats, NEXUS turns every scan into a learning opportunity and a stronger defense.

CONTACT INFORMATION – Team Opulence

For demonstrations, partnership opportunities, or employment inquiries, please contact any team member via LinkedIn or email.

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Appendix A: Technical Glossary

The following terms are referenced throughout this memo and the accompanying presentation slides.

Term	Definition
API (Application Programming Interface)	A set of rules that allows different software applications to communicate. NEXUS uses RESTful APIs to connect the frontend to the backend.
AVG	An antivirus tool offering fast signature-based scanning. While effective for known threats, it has limited explainability and may miss new or obfuscated malware.
AWS (Amazon Web Services)	A cloud computing platform providing on-demand services. NEXUS uses AWS for future email automation capabilities.
CSS (Cascading Style Sheets)	A styling language used to control the visual presentation of web pages, including layout, colors, and responsive design.
EML	A standard file format for email messages. NEXUS analyzes .eml files to extract headers, metadata, attachments, and phishing indicators.
Flask	A lightweight Python web framework used to build the NEXUS backend API layer, handling HTTP routing, file uploads, and response formatting.
IOC (Indicator of Compromise)	Observable artifacts such as IP addresses, URLs, file hashes, or registry keys that indicate a system may have been breached.
OCR (Optical Character Recognition)	Technology that converts images of text into machine-readable text. NEXUS uses Tesseract OCR to detect hidden text in image files.
PyPDF2	A Python library for reading and extracting text from PDF files. Used by NEXUS to process PDF uploads for analysis.
python-docx	A Python library for reading content from Microsoft Word (.docx) files, enabling NEXUS to analyze Word document attachments.
React.js	A JavaScript library for building user interfaces through reusable components. NEXUS uses React.js for the entire frontend application.
Regex (Regular Expressions)	Pattern-matching syntax used to identify structured text such as IP addresses, URLs, email addresses, and suspicious keywords within file content.
RESTful API	An architectural style for web services where clients and servers communicate using standard HTTP methods (GET, POST) with stateless interactions.
Snowflake	A cloud-based data warehousing platform used by NEXUS to store scan history, analysis metadata, and historical threat data with 30-day retention.
Static Analysis	A method of examining files without executing them. NEXUS reads file content, metadata, and structure to detect threats without running any embedded code.
Steganography	The practice of hiding information within other data, such as concealing malicious text within image pixels. NEXUS's OCR engine can detect some forms of this.
Tesseract	An open-source OCR engine supporting 100+ languages. Integrated into NEXUS to extract text from image files (PNG, JPEG) for indicator scanning.
VirusTotal	A web-based tool that aggregates multiple antivirus engines for quick file triage. Useful for rapid scanning but requires uploading files to external servers.
WCAG 2.1 AA	Web Content Accessibility Guidelines version 2.1 at the AA level. Ensures web applications are usable by people with disabilities, including screen reader users.
WebSocket	A communication protocol providing full-duplex, real-time data transfer between client and server. NEXUS uses WebSockets for live progress updates during analysis.